

EMPLOYEE BURNOUT EARLY DETECTION SYSTEM

A MINI PROJECT REPORT

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BONAFIDE CERTIFICATE
EMPLOYEE BURNOUT EARLY DETECTION SYSTEM

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ABSTRACT

In large corporate environments, manually monitoring employee performance and identifying non-productive resources is a complex and time-consuming task. Traditional Human Resource (HR) evaluations often suffer from subjectivity and human error, leading to the retention of underperforming staff or efficient employees being overlooked. This project proposes an automated **Employee Burnout Early Detection System** designed to assist management in making data-driven decisions regarding workforce optimization and involuntary attrition. The proposed system utilizes Machine Learning algorithms, specifically Classification techniques (e.g., Decision Tree / Logistic Regression), to analyze key performance indicators (KPIs). The model is trained on critical behavioral and productivity metrics, including **project submission delays, frequency of absenteeism, number of behavioral complaints, and average effective working hours**. By processing these parameters, the system generates a prediction score that classifies employees into "Retain," "Review," or "Terminate" categories. The primary objective is to provide the management team with a transparent, rule-based mechanism to identify employees who consistently fail to meet company standards. This solution reduces the manual workload for HR professionals, ensures objective evaluation based on factual data logs, and helps the organization maintain high operational efficiency by filtering out non-performing assets.

LIST OF ABBREVIATIONS

S.NO	ABBREVIATIONS	EXPANSION
1	LR	Linear Regression
2	SVM	Super Vector Machine
3	RF	Random Forest
4	GBM	Gredient Boosting Machine
5	DF	Data Frame
6	RDBMS	Relational Data Base Management System
7	Sk-Learn	Scikit Learn
8	SNS	Seaborn
9	PD	Pandas
10	CSV	Comma Separated Value
11	KPI	Key Performance Indicator

12	HR	Human Resources
13	UI	User Interface
14	API	Application Programming Interface
15	ML	Machine Learning

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CHAPTER 1 – INTRODUCTION

1.1 About the Project

The Employee Burnout Early Detection System is an advanced analytical platform developed to monitor employee productivity, wellness, attendance, and behavioral consistency within a corporate environment. The primary objective of this system is to identify signs of employee burnout and poor performance at an early stage through automated analysis and intelligent data processing.

In modern organizations, employee burnout has become one of the most critical challenges faced by Human Resource departments. Increasing workloads, tight deadlines, long working hours, workplace pressure, stress, and unhealthy work-life balance significantly affect employee mental health and productivity. Burnout not only reduces the efficiency of employees but also impacts the overall growth and profitability of an organization.

Traditional methods of employee monitoring are mostly manual and rely heavily on supervisors or managers to evaluate employee performance. These methods are often slow, inaccurate, and influenced by personal opinions or favoritism. In large organizations with hundreds or thousands of employees, it becomes extremely difficult for HR teams to continuously monitor every employee effectively.

To overcome these limitations, the Employee Burnout Early Detection System introduces a Machine Learning-based approach that automates employee performance and wellness evaluation. The system collects and analyzes multiple employee-related parameters such

as attendance records, overtime hours, absenteeism, late task submissions, complaints, productivity rates, and behavioral patterns. By processing this data, the system predicts whether an employee is healthy, at risk, or in a critical burnout stage.

Machine Learning algorithms play a vital role in the project by identifying hidden patterns and trends within employee data. The system continuously learns from historical data and improves prediction accuracy over time. This helps organizations make faster and more reliable decisions regarding employee wellness and performance management.

The project interface is designed using modern web technologies such as HTML, CSS, JavaScript, and Tailwind CSS. The dashboard provides an attractive, responsive, and user-friendly environment for HR teams and administrators. Various visual indicators including employee wellness scores, burnout risk levels, alerts, performance statistics, and analytical charts are displayed in real time. These features allow HR managers to quickly identify employees who require attention or support.

Another important advantage of the system is transparency in evaluation. Since the analysis is based on data-driven insights rather than personal opinions, the chances of bias and unfair judgment are greatly reduced. This increases employee trust in the evaluation process and improves workplace fairness.

The proposed system also helps organizations save valuable time and resources. Instead of manually checking employee records and performance reports, HR teams can depend on automated analysis for faster decision-making. The scalable architecture of the system makes it suitable for both small businesses and large enterprises.

Overall, the Employee Burnout Early Detection System provides an intelligent, efficient, and scalable solution for employee monitoring and wellness management. By detecting burnout at an early stage, organizations can improve employee satisfaction, reduce productivity loss, enhance workforce optimization, and create a healthier working environment.

1.2 Goals of Project

The Employee Burnout Early Detection System is developed with the aim of improving employee wellness monitoring and organizational productivity through automation and Machine Learning technologies. The project focuses on identifying employee stress and burnout symptoms at an early stage so that organizations can take preventive actions before performance decreases significantly.

The major goals of the project are explained below:

1. To automate employee wellness monitoring

The project aims to replace traditional manual monitoring methods with an automated system that continuously analyzes employee activities, attendance, productivity, and behavior. Automation reduces the workload of HR departments and increases efficiency.

2. To detect burnout at an early stage

One of the primary objectives of the system is to identify signs of stress, exhaustion, and reduced productivity before employees reach a critical burnout stage. Early detection helps organizations provide timely support and corrective measures.

3. To improve decision-making using Machine Learning

The system uses Machine Learning algorithms to analyze employee data and generate accurate predictions. These predictions help HR managers make informed decisions regarding employee performance, workload distribution, and wellness management.

4. To reduce manual HR workload

Managing employee performance manually requires significant time and effort. The proposed system automates data analysis and report generation, thereby reducing the burden on HR professionals.

5. To identify high-risk employees quickly

The project helps organizations quickly identify employees who are experiencing high stress levels, excessive workload, or reduced productivity. This enables faster intervention and better workforce management.

6. To provide accurate performance analysis

The system evaluates employees based on real-time data rather than subjective opinions. This improves the accuracy and reliability of performance analysis.

7. To create a transparent evaluation process

A data-driven system ensures fairness and transparency in employee evaluation. Employees are assessed using measurable parameters, reducing favoritism and bias.

8. To reduce organizational productivity loss

Burnout often leads to absenteeism, reduced efficiency, low morale, and employee

turnover. By detecting burnout early, organizations can minimize productivity loss and improve overall operational performance.

9. To improve workforce optimization

The project helps organizations optimize workforce management by identifying productivity trends, workload distribution issues, and employee wellness conditions.

10. To develop an interactive dashboard for HR management

The system includes a modern and interactive dashboard designed using HTML, CSS, JavaScript, and Tailwind CSS. The dashboard displays employee risk levels, wellness scores, alerts, and performance statistics in an easy-to-understand format.

The successful implementation of these goals will help organizations maintain a healthier work environment, improve employee satisfaction, and achieve better productivity outcomes.

1.3 Problem Statement

In today's corporate environment, organizations face major challenges in managing employee wellness and maintaining consistent productivity. Employee burnout has become increasingly common due to work pressure, excessive responsibilities, long working hours, and stressful organizational environments. Burnout negatively affects both employees and companies by reducing efficiency, increasing absenteeism, and lowering overall workplace morale.

Traditional employee monitoring systems mainly depend on manual evaluations performed

by managers and HR departments. These systems are inefficient because they rely heavily on human observation, subjective judgment, and periodic reviews rather than continuous monitoring. As a result, organizations often fail to identify employee stress and burnout at the right time.

Several major problems exist in traditional employee evaluation systems:

- Favoritism and bias during performance evaluation
- Manual evaluations are often influenced by personal opinions, relationships, or favoritism. This creates unfair performance assessments and reduces employee trust in the organization.
- Difficulty in tracking employee activities daily
- Monitoring employee attendance, productivity, task completion, overtime, and behavioral changes manually is difficult, especially in large organizations with many employees.
- Lack of real-time monitoring
- Traditional systems do not provide real-time insights into employee performance and wellness. Problems are often identified only after productivity has already decreased.
- Delays in decision-making
- Since HR departments need to collect and analyze large amounts of data manually, decision-making becomes slow and inefficient.
- Poor detection of employee stress and burnout

- Human observation alone is not sufficient to identify hidden stress patterns or emotional exhaustion. Employees may continue working under pressure until burnout becomes severe.
- Inefficient data handling
- Managing employee records manually increases complexity and consumes significant organizational resources. Data may also become inconsistent or difficult to analyze
- High chances of human error
- Manual systems are more prone to mistakes during data entry, evaluation, and report preparation. Such errors can affect important HR decisions.
- To address these problems, the Employee Burnout Early Detection System introduces a Machine Learning-based solution that automates employee analysis and monitoring. The system collects employee-related data such as attendance records, overtime details, late submissions, complaints, and productivity metrics. Based on this information, Machine Learning algorithms analyze employee behavior and predict burnout risk levels.
- The system provides real-time monitoring, automated alerts, and accurate performance analysis through an interactive dashboard. HR managers can easily identify employees who are healthy, at risk, or in critical condition. This helps organizations take preventive measures before burnout affects employee

performance and organizational productivity.

- By reducing manual intervention and improving analytical accuracy, the proposed system provides a reliable, transparent, and scalable solution for employee wellness management. The project ultimately aims to create a healthier work environment, improve employee satisfaction, and enhance overall organizational efficiency.

CHAPTER 2 – LITERATURE SURVEY

2.1 Machine Learning in HR Analytics

Machine Learning (ML) has significantly transformed the field of Human Resource Management (HRM) by introducing intelligent systems capable of analyzing large volumes of employee-related data (IBM Machine Learning Fundamentals, Human Resource Analytics Research). Traditional HR systems mainly depended on manual observation, spreadsheets, and basic reports, which often failed to identify hidden behavioral patterns and future risks. Modern HR analytics powered by Machine Learning enables organizations to make accurate, data-driven decisions.

HR analytics refers to the process of collecting, analyzing, and interpreting employee data to improve workforce performance and organizational productivity (Artificial Intelligence in Workforce Management, Employee Wellness Research Articles). Companies now use Machine Learning algorithms to analyze employee attendance, work efficiency, stress levels, engagement scores, overtime patterns, and communication behavior.

Employee burnout has become one of the major challenges faced by organizations worldwide. Continuous workload pressure, stress, lack of work-life balance, and unhealthy work environments negatively affect employee performance and mental wellness. Machine Learning helps HR departments detect early burnout symptoms before they become severe (Human Resource Analytics Research, Employee Wellness Research Articles).

Organizations increasingly use Machine Learning to:

- Predict employee attrition.
- Analyze employee behavior.
- Improve recruitment and hiring.
- Detect productivity trends.
- Monitor employee performance.
- Identify burnout symptoms.
- Improve employee satisfaction.
- Reduce organizational losses.

The integration of Machine Learning in HR provides several advantages such as :

- faster decision-making
- improved workforce management
- better prediction accuracy
- reduced manual effort
- automated employee monitoring
- enhanced employee engagement (Predictive Analytics Overview, Data Science and Analytics Tutorials).

Machine Learning systems can continuously learn from historical employee data and improve prediction performance over time. HR departments can use predictive analytics to

identify employees who may face stress, dissatisfaction, or performance decline (IBM Machine Learning Fundamentals, Predictive Analytics Overview).

Applications of Machine Learning in HR

1. Recruitment Analytics

- Automatically filters resumes.
- Predicts suitable candidates.
- Reduces hiring time.

2. Employee Performance Analysis

- Measures employee productivity.
- Tracks attendance and task completion.
- Identifies performance trends.

3. Burnout Detection

- Detects stress indicators.
- Monitors work overload.
- Predicts employee risk levels.

4. Employee Retention

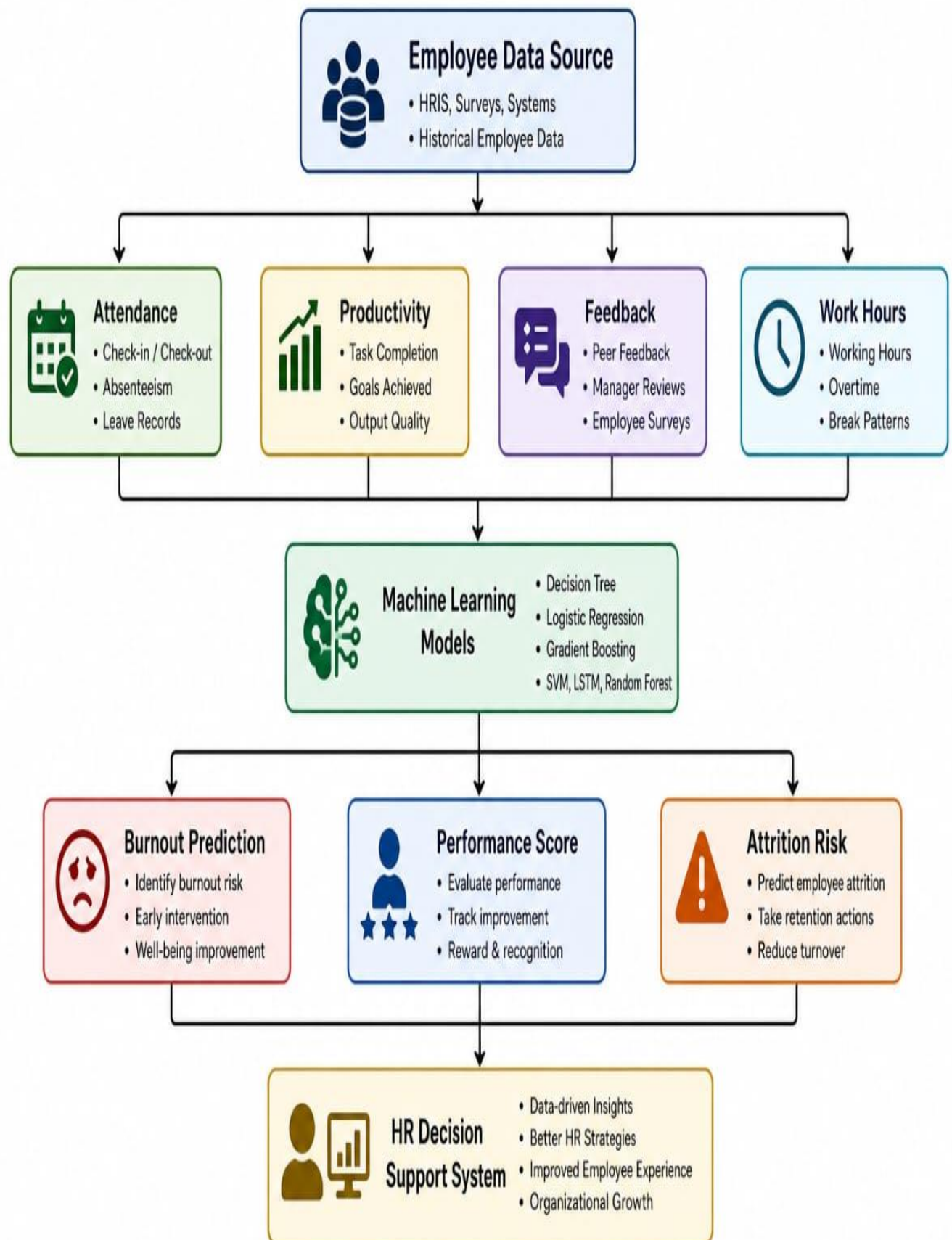
- Predicts employee resignation probability.
- Improves employee satisfaction strategies.

5. Workforce Planning

- Assists HR managers in planning workforce requirements.
- Optimizes organizational efficiency.

Diagram:

Machine Learning in HR Analytics



2.2 Decision Tree Model

A Decision Tree is a supervised Machine Learning algorithm used for both classification and regression tasks (Scikit Learn Documentation, Machine Learning Course Materials). It divides the dataset into smaller subsets based on feature conditions and creates a tree-like structure consisting of nodes, branches, and leaf nodes.

The root node represents the entire dataset, while internal nodes represent decision conditions. Leaf nodes indicate the final prediction or classification result. Decision Trees are highly preferred in HR analytics because they are simple to understand and provide visual interpretations of employee classification results (Machine Learning Fundamentals – IBM).

In the Employee Burnout Detection System, Decision Trees classify employees into categories such as:

- Healthy Employee
- Moderate Risk Employee
- High Burnout Risk Employee

The algorithm analyzes features such as:

- Working hours
- Overtime frequency
- Attendance records
- Employee engagement

- Productivity score
- Stress indicators

Working Process of Decision Tree

1. Select the best feature from the dataset.
2. Split the dataset based on conditions.
3. Create branches for each outcome.
4. Repeat recursively until prediction is obtained.

Advantages

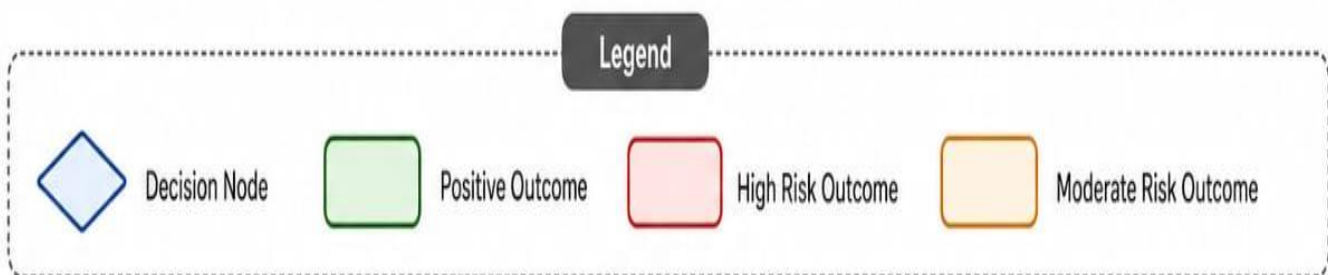
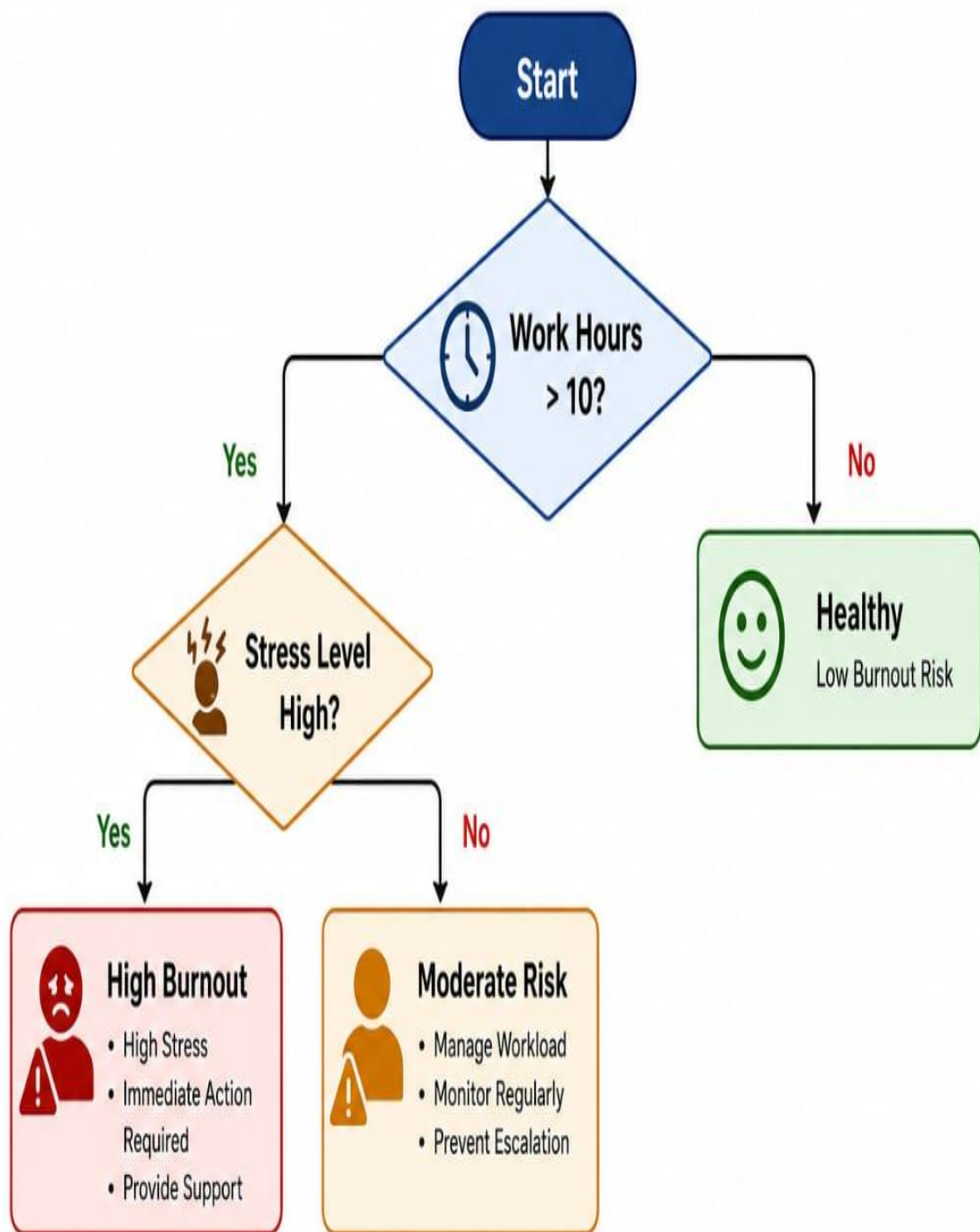
- Easy to understand.
- Provides visual representation.
- Fast prediction.
- Handles numerical and categorical data.
- Requires less data preprocessing.

Disadvantages

- Can overfit data.
- Sensitive to noisy datasets.
- Accuracy may reduce with complex datasets.

Diagram:

Decision Tree Structure



2.3 Logistic Regression

Logistic Regression is a statistical classification algorithm widely used in Machine Learning for binary classification problems (IBM Machine Learning Fundamentals, Predictive Analytics Overview). It predicts the probability of an outcome occurring using a logistic function.

In HR analytics, Logistic Regression helps determine whether an employee belongs to a safe category or burnout-risk category. The algorithm calculates probabilities between 0 and 1. If the probability exceeds a threshold value, the employee is classified as burnout risk.

For example:

- Probability $> 0.5 \rightarrow$ Burnout Risk
- Probability $< 0.5 \rightarrow$ Healthy Employee

Logistic Regression is effective for analyzing linear relationships between employee attributes and burnout status (Machine Learning Course Materials).

Applications in HR Analytics

- Employee attrition prediction.
- Burnout classification.
- Attendance analysis.
- Employee satisfaction analysis.

Advantages

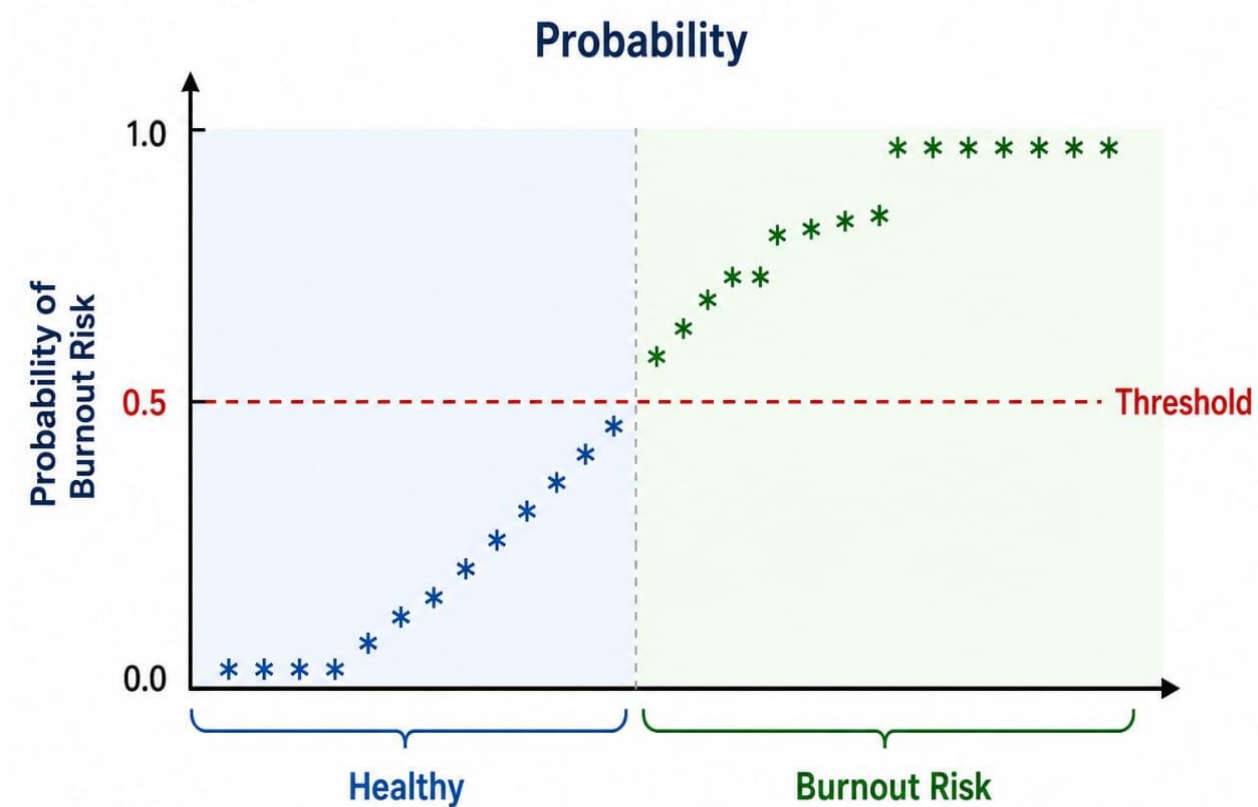
- Simple implementation.
- High efficiency.
- Good accuracy for linear datasets.
- Faster training process.
- Easy interpretation.

Disadvantages

- Limited performance on complex data.
- Sensitive to outliers.
- Requires feature scaling.

Diagram:

Logistic Regression Classification



2.4 Gradient Boosting Machines

Gradient Boosting Machine (GBM) is an advanced ensemble Machine Learning algorithm that combines multiple weak learning models to create a powerful predictive model (Data Science and Analytics Tutorials, IBM Machine Learning Fundamentals). The algorithm works by building models sequentially.

Each new model attempts to correct the errors made by the previous model. This iterative improvement process increases overall prediction accuracy. In HR analytics, Gradient Boosting is useful for analyzing complex employee behavior patterns and identifying hidden burnout indicators.

GBM performs exceptionally well when datasets contain multiple influencing factors such as:

- Stress level
- Workload
- Overtime hours
- Employee feedback
- Project pressure
- Performance trends

Working Principle

1. Train the first weak model.
2. Measure prediction error.
3. Train another model to correct previous errors.
4. Repeat until optimal performance is achieved.

Advantages

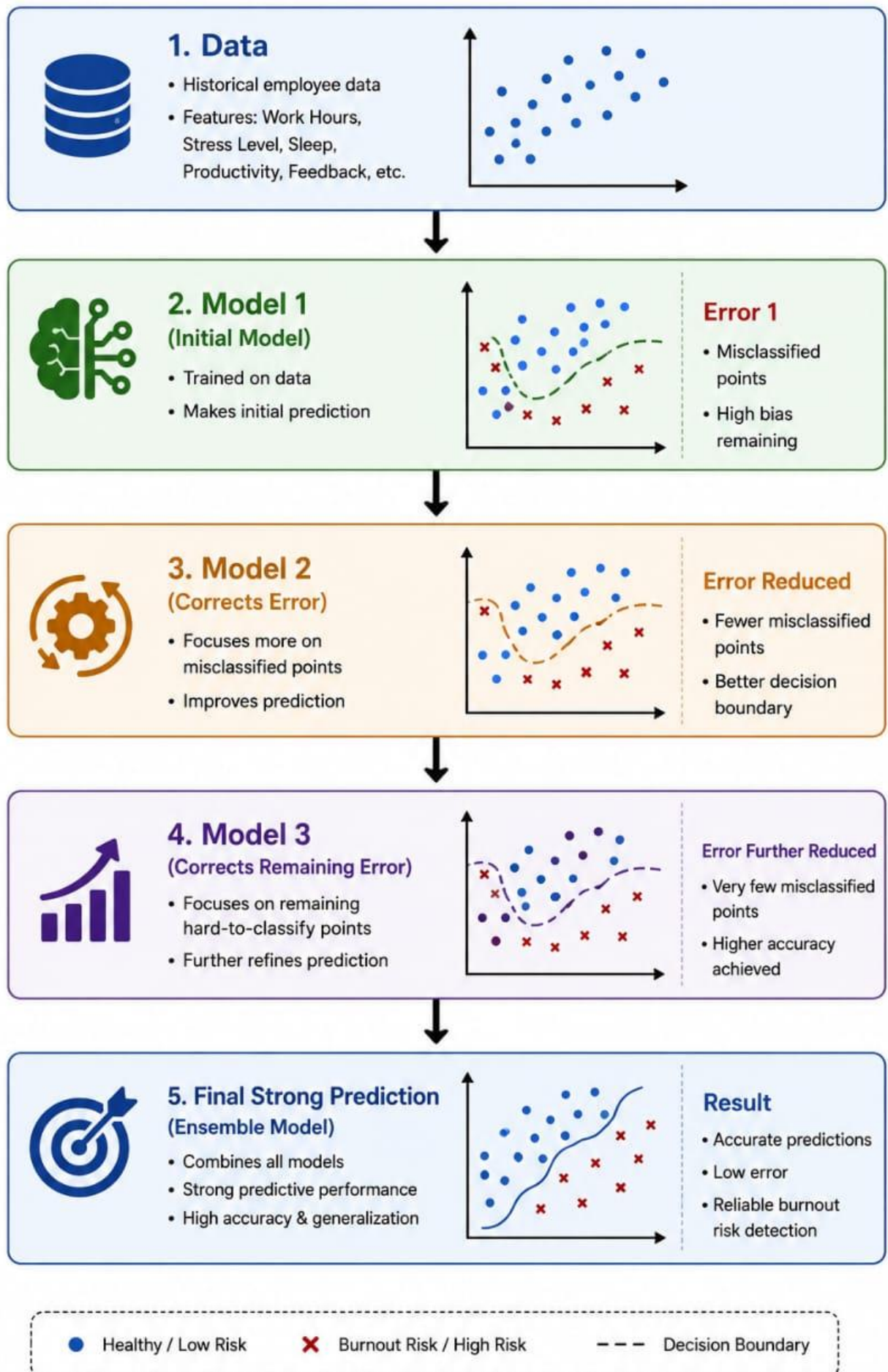
- High predictive accuracy.
- Handles complex relationships.
- Reduces prediction errors.
- Works well with large datasets.

Disadvantages

- Computationally expensive.
- Requires parameter tuning.
- Slower training process.

Diagram:

Gradient Boosting Process



2.5 Support Vector Machine

Overtime hours Support Vector Machine (SVM) is a supervised Machine Learning algorithm mainly used for classification tasks (Scikit Learn Documentation, Machine Learning Course Materials). It separates data into categories using hyperplanes.

The primary objective of SVM is to identify the optimal boundary that maximizes the distance between different classes. In employee burnout prediction systems, SVM classifies employees based on work behavior and performance features.

Features Used in SVM

- Employee workload
- Stress level
- Attendance
- Performance score

Advantages

- Effective in high-dimensional datasets.
- High classification accuracy.
- Works well with smaller datasets.

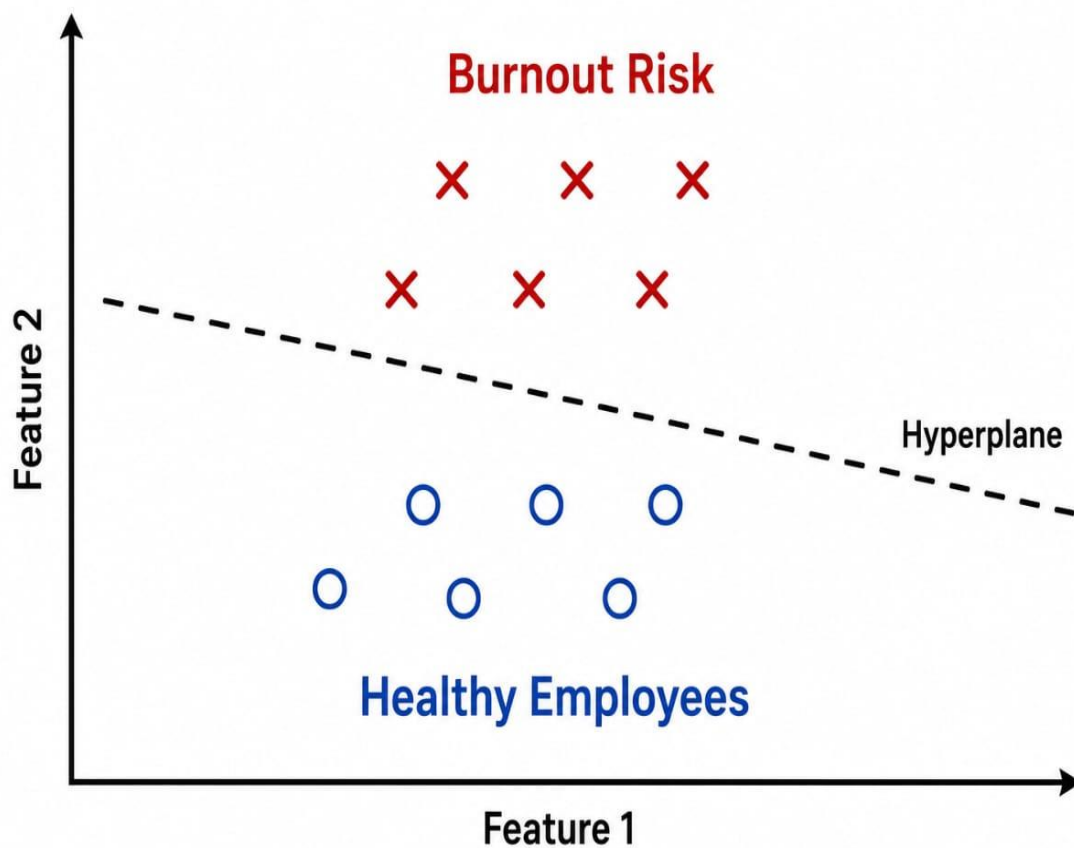
Disadvantages

- Slower on large datasets.
- Difficult parameter tuning.

- Requires careful kernel selection.

Diagram:

Support Vector Machine



2.6 Long Short-Term Memory Networks

Long Short-Term Memory (LSTM) is a Deep Learning algorithm that belongs to the Recurrent Neural Network (RNN) family (Data Science and Analytics Tutorials, Machine

Learning Course Materials). LSTM is specially designed to process sequential and time-series data.

Unlike traditional Machine Learning models, LSTM can remember long-term dependencies and patterns in data. In employee analytics, LSTM helps monitor employee behavior over long periods and identify gradual burnout symptoms.

Examples of sequential employee data include:

- Daily attendance records.
- Monthly productivity scores.
- Stress level history.
- Workload variations.
- Task completion trends.

LSTM models contain memory cells that store important information and forget unnecessary details.

Advantages

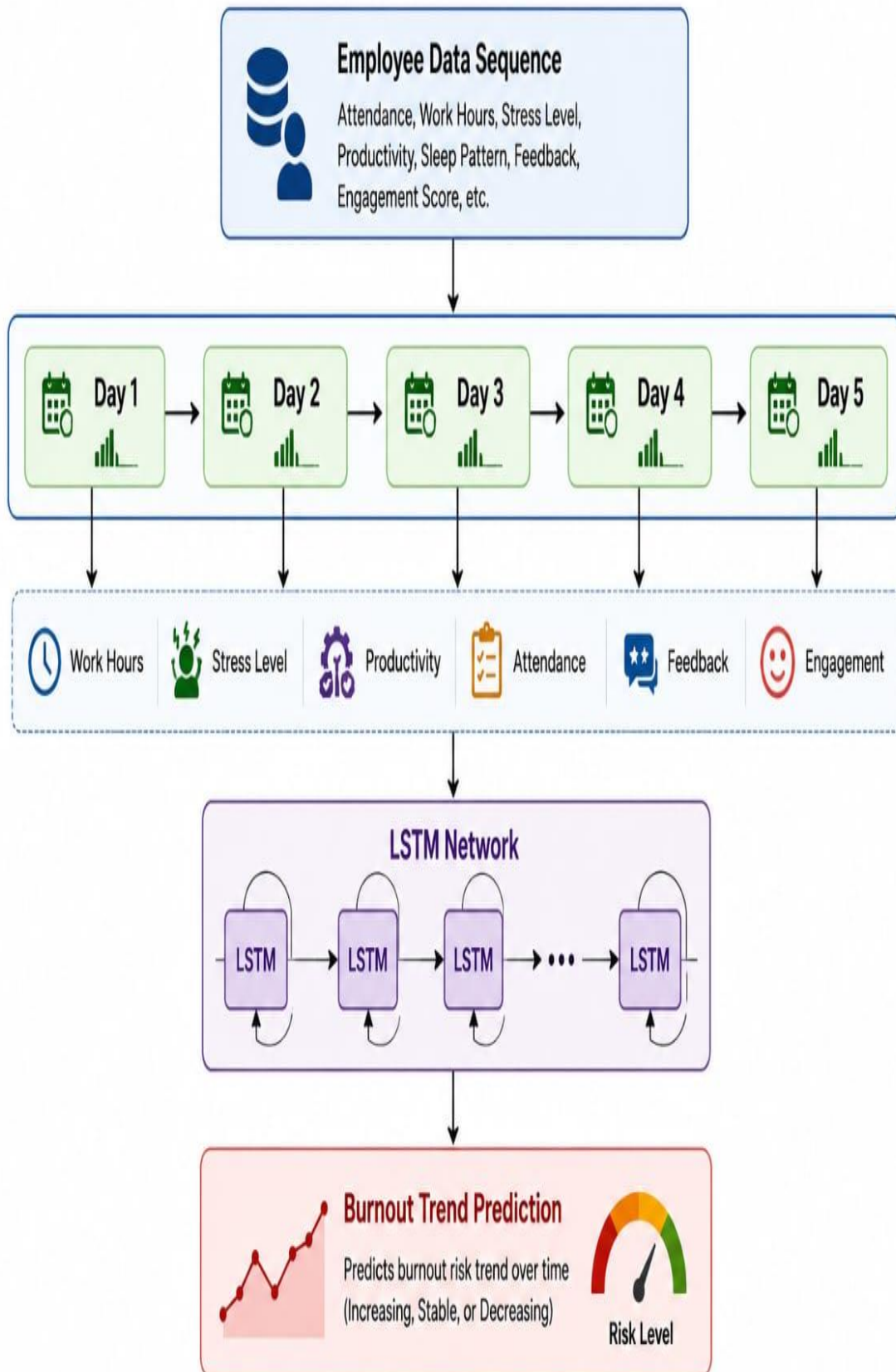
- Handles sequential data effectively.
- Learns long-term behavior patterns.
- Suitable for time-series prediction.
- High prediction capability.

Disadvantages

- Requires high computational power.
- Complex implementation.
- Long training time.

Diagram:

LSTM Workflow



2.7 Random Forest

Random Forest is an ensemble Machine Learning algorithm that combines multiple Decision Trees to improve prediction accuracy and reduce overfitting (Scikit Learn Documentation, IBM Machine Learning Fundamentals). Each Decision Tree is trained using random subsets of data and features.

The final prediction is obtained through majority voting. Random Forest is widely used in HR analytics because it provides stable and reliable predictions even when datasets contain missing or noisy values.

In Employee Burnout Detection Systems, Random Forest analyzes multiple employee-related factors simultaneously and predicts burnout risk effectively.

Working Principle

1. Create multiple Decision Trees.
2. Train each tree with random data subsets.
3. Collect predictions from all trees.
4. Final prediction is selected using majority voting.

Advantages

- High accuracy.
- Reduced overfitting.

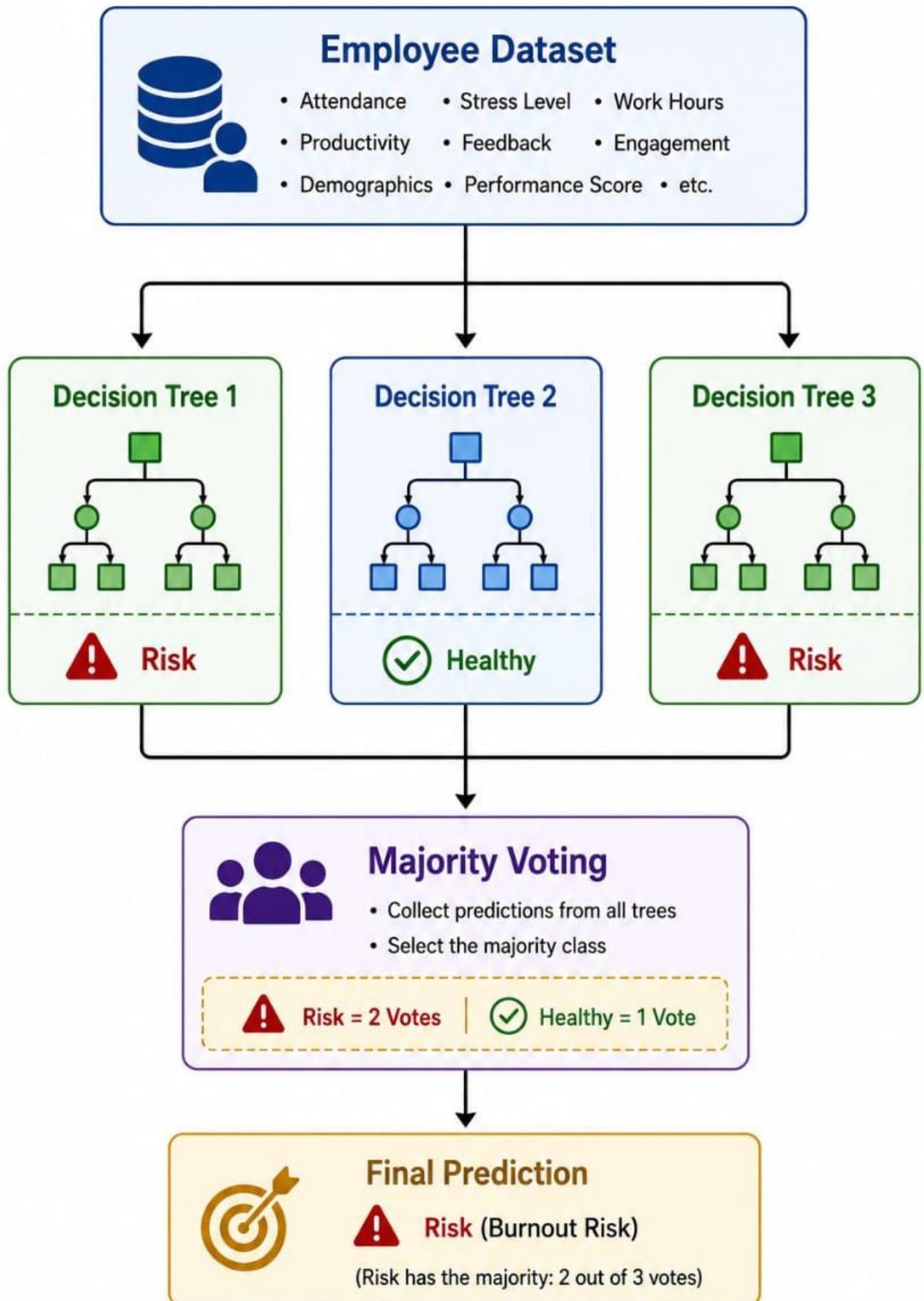
- Handles missing values.
- Works with large datasets.
- Strong prediction stability.

Disadvantages

- Slower than single Decision Trees.
- Complex interpretation.
- Requires more memory.

Diagram:

Random Forest Architecture



CHAPTER 3 – EXISTING SYSTEM

Currently, many companies rely on manual performance reviews conducted by managers. Traditional Human Resource Management systems mainly depend on employee observation, supervisor feedback, attendance records, and periodic appraisal methods to evaluate employee performance. These systems are widely used in organizations because they are simple to implement and easy to understand.

In the existing system, employee data such as attendance, productivity, task completion, overtime details, and feedback are collected and reviewed by HR teams or managers. Performance evaluations are usually conducted monthly, quarterly, or annually. Based on these evaluations, organizations make decisions regarding promotions, incentives, workload distribution, and employee support.

Traditional systems are effective for maintaining employee records and managing organizational activities. They help companies monitor employee attendance, maintain productivity reports, and support administrative decision-making processes. Managers also use these systems to understand employee performance trends and workplace behavior.

The existing system mainly focuses on historical performance records and manager observations. Employee activities are reviewed using reports, spreadsheets, and manual analysis methods. This approach provides organizations with structured employee information and supports workforce management processes.

Features of Existing System

1. Manager-Based Evaluation

Managers and supervisors evaluate employee performance based on work quality, attendance, behavior, and productivity. Their experience helps organizations understand employee strengths and workplace performance.

2. Periodic Performance Monitoring

The system allows organizations to review employee performance at regular intervals such as monthly, quarterly, or yearly assessments. This helps maintain employee records and monitor organizational progress.

3. Employee Activity Tracking

Organizations track important employee activities including:

- Attendance
- Work hours
- Task completion
- Productivity levels
- Feedback reports

This information supports employee management and performance evaluation.

4. Report Generation

HR departments generate employee performance reports using collected organizational

data. These reports help management in decision-making processes and workforce planning.

5. HR Management Support

The system supports HR teams in maintaining employee records, monitoring attendance, evaluating performance, and handling organizational operations efficiently.

6. Employee Performance Analysis

Managers analyze employee productivity and engagement levels using collected performance records. This helps organizations identify active employees and improve workforce management.

7. Workplace Monitoring

The system helps organizations monitor employee discipline, punctuality, teamwork, and work consistency through regular evaluations and observations.

8. Organizational Decision Support

Performance evaluations assist organizations in making important decisions related to promotions, rewards, training programs, and employee development activities.

The traditional approach provides a structured method for managing employee information and organizational activities. It serves as the foundation for employee performance management and supports HR departments in maintaining workforce records and operational efficiency

CHAPTER 4 – PROPOSED SYSTEM

The proposed system automates the employee monitoring process using a centralized database and Machine Learning models.

Advantages of Proposed System

1. Data-driven evaluation.
2. Real-time monitoring.
3. Early warning alerts.
4. Reduced manual work.
5. Faster HR decisions.
6. Better workforce management.
7. Transparent evaluation.
8. Scalable architecture.
9. Improved employee wellness tracking.
10. Smart analytical dashboard.

The system predicts employee burnout using productivity metrics and generates alerts for management.

CHAPTER 5 – SYSTEM REQUIREMENTS

5.1 Software Requirements

Software	Description
HTML	Structure of web pages
CSS	Styling and layout
JavaScript	Dynamic functionality
Tailwind	Responsive UI framework
VS Code	Development environment
Browser	Running the application

5.2 Hardware Requirements.

Hardware	Specification
Processor	Intel i3 or above
RAM	4 GB minimum
Storage	500 GB
Monitor	LED Display
Internet	Required

5.3 Software Description

HTML

HTML is used to design the structure of the application.

CSS

CSS is used for styling and creating responsive layouts.

JavaScript

JavaScript handles interactivity and dynamic calculations.

Tailwind CSS

Tailwind CSS provides utility-based styling for modern UI design.

CHAPTER 6 – SYSTEM DESIGN

6.1 System Architecture

System architecture defines the overall structure and workflow of the Employee Burnout Early Detection System. The proposed system follows a layered architecture model where each layer performs a specific function. This design improves scalability, efficiency, security, and maintainability of the application.

The architecture of the system contains five important layers:

1. User Interface Layer
2. Data Processing Layer
3. Machine Learning Prediction Layer
4. Visualization Layer
5. Alert Generation Layer

The interaction between these layers allows the system to monitor employee performance, analyze productivity metrics, predict burnout conditions, and display analytical reports in real time.

User Interface Layer

The User Interface Layer is the front-end component of the system that directly interacts with users such as HR Managers and System Administrators. The interface is designed using HTML, CSS, JavaScript, and Tailwind CSS.

This layer provides a modern dashboard where management can monitor employee wellness and organizational productivity.

Functions of User Interface Layer

- Display employee dashboard
- Show employee wellness score
- Display attendance reports
- Generate visual charts and tables
- Display employee alerts
- Show healthy, risk, and critical employee categories
- Allow filtering and searching of employee data

The user interface is responsive, attractive, and easy to use. Different color indicators are used to represent employee conditions:

- Green → Healthy
- Yellow → At Risk
- Red → Critical

Data Processing Layer

The Data Processing Layer is responsible for collecting and preprocessing employee-related information. Raw employee data may contain incomplete values, inconsistent

formats, or duplicate records. Therefore, preprocessing is essential before analysis.

Employee Data Used

The system collects various employee performance metrics such as:

- Attendance records
- Overtime hours
- Project submission delays
- Number of complaints
- Task completion rate
- Average working hours
- Productivity score

Functions of Data Processing Layer

- Data collection
- Data cleaning
- Data normalization
- Feature extraction
- Data transformation

The preprocessing stage improves prediction accuracy and ensures reliable Machine Learning performance.

Machine Learning Prediction Layer

The Machine Learning Prediction Layer is the core component of the proposed system. This layer analyzes employee data and predicts burnout conditions using classification algorithms.

The system evaluates employee performance metrics and calculates a burnout score. Based on the score, employees are categorized into different risk levels.

Algorithms Used

The proposed system uses Machine Learning classification algorithms such as:

- Decision Tree
- Logistic Regression
- Random Forest

These algorithms analyze employee behavior patterns and identify employees who may experience burnout or low productivity.

Classification Categories

Employees are classified into three categories:

1. Healthy
2. At Risk
3. Critical

Functions of Machine Learning Layer

- Predict burnout score
- Analyze productivity
- Detect stress conditions
- Classify employee performance
- Generate analytical results

This layer improves organizational decision-making by providing objective evaluations based on employee data.

Visualization Layer

The Visualization Layer converts prediction results into graphical representations and dashboard analytics.

The visualization process helps HR teams understand employee conditions easily using charts, graphs, and tables.

Functions of Visualization Layer

- Display employee statistics
- Generate performance charts
- Display risk analysis reports
- Show employee wellness trends
- Present graphical analytics

The dashboard provides real-time monitoring features and visual representations of employee performance.

Alert Generation Layer

The Alert Generation Layer continuously monitors burnout scores and generates warnings when employees enter critical conditions.

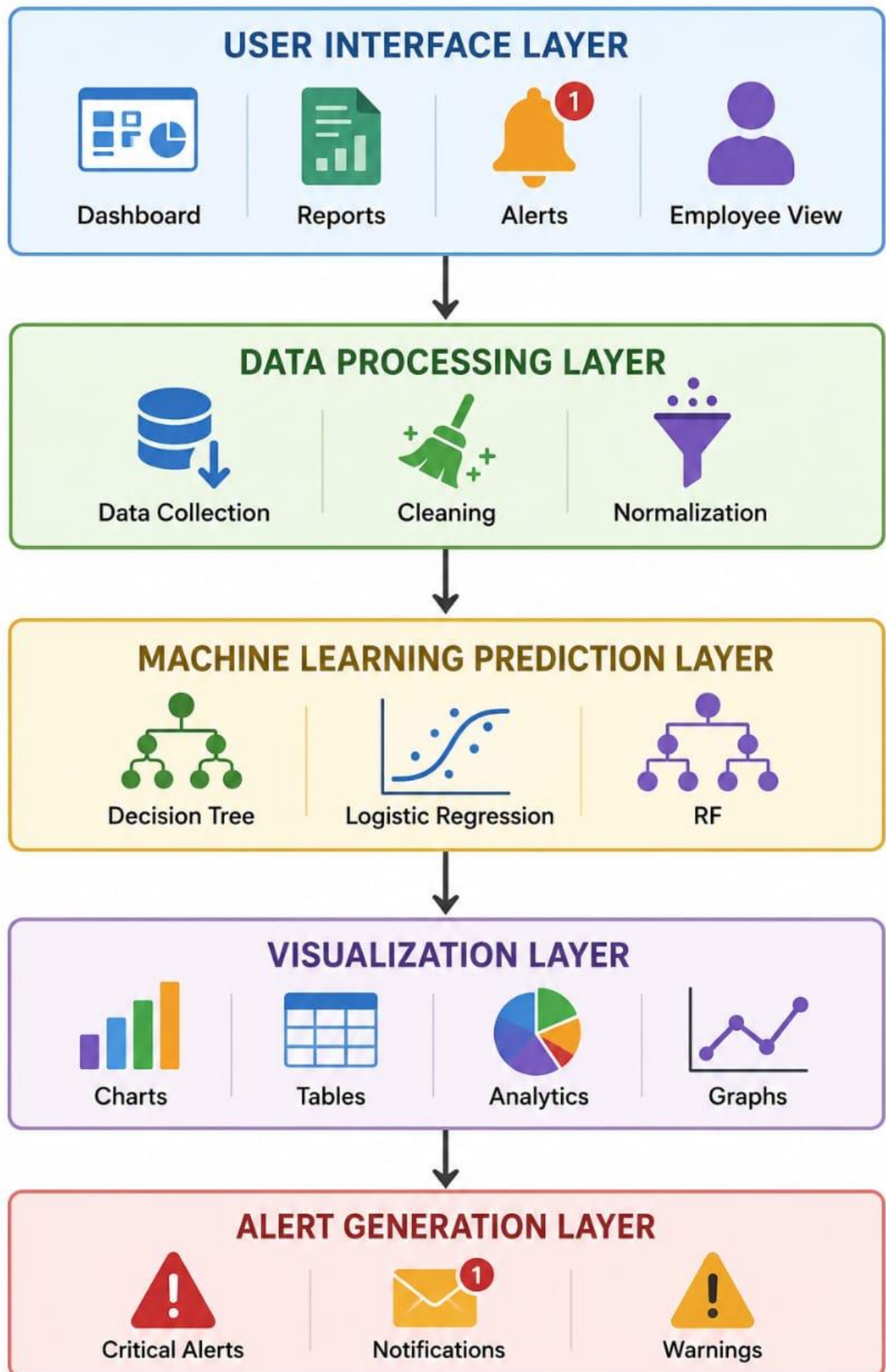
This layer helps organizations take immediate action before employee burnout affects company productivity.

Functions of Alert Generation Layer

- Generate real-time alerts
- Detect critical employees
- Notify HR management
- Display emergency warnings
- Provide risk notifications

This feature supports early intervention and workforce optimization.

System Architecture Diagram



Working Process

The Employee Burnout Early Detection System follows a step-by-step workflow for employee monitoring and prediction.

Step 1: Employee Data Collection

Employee-related information such as attendance, overtime, complaints, and productivity is collected from organizational records.

Step 2: Data Processing and Normalization

The collected data is cleaned and normalized to improve consistency and remove invalid entries.

Step 3: Feature Extraction

Important performance indicators are extracted for Machine Learning analysis.

Step 4: Burnout Score Calculation

The Machine Learning model calculates burnout scores based on employee performance metrics.

Step 5: Employee Classification

Employees are categorized into Healthy, At Risk, or Critical groups according to the prediction score.

Step 6: Dashboard Visualization

The final analytical results are displayed on the dashboard using graphs, charts, and tables.

Step 7: Alert Generation

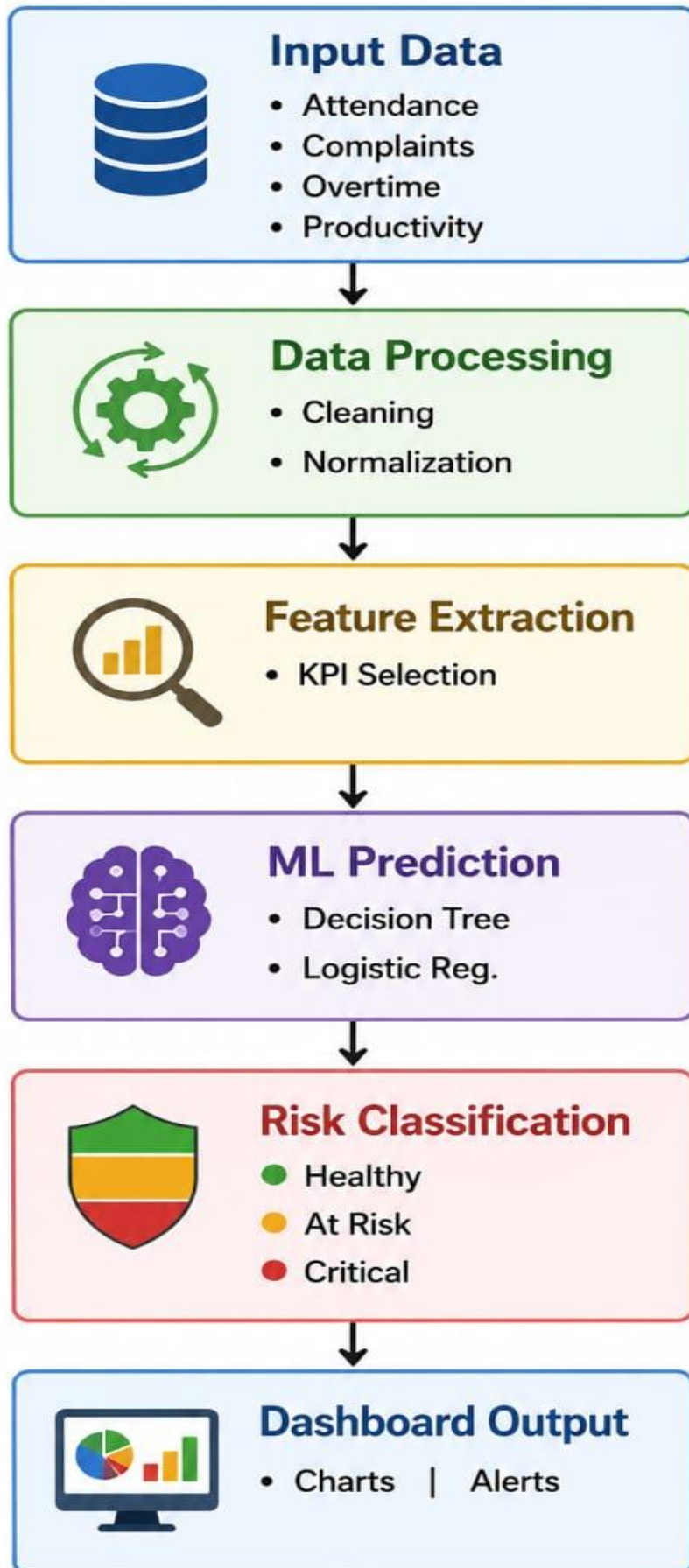
Real-time notifications and alerts are generated for critical employees.

6.2 Model Flow Chart

The model flow chart explains the complete processing flow of the Employee Burnout Early Detection System.

The process starts with employee data collection and ends with prediction visualization and alert generation.

Model Flow Chart



Explanation of Flow Chart

Input Stage

Employee performance metrics are collected from the database.

Data Processing Stage

The collected data is cleaned and normalized.

Feature Extraction Stage

Important performance indicators are selected for prediction.

Machine Learning Prediction Stage

Classification algorithms analyze employee performance.

Classification Stage

Employees are grouped into Healthy, At Risk, or Critical categories.

Output Stage

The final prediction results are displayed on the dashboard.

6.3 System Use Case

A Use Case Diagram describes the interaction between users and the system.

The proposed system contains three main actors:

- HR Manager
- System Administrator

- Employee
- Actors in the System

1. HR Manager

The HR Manager monitors employee wellness and organizational productivity.

Responsibilities:

- Monitor employee performance
- View burnout analysis
- Generate reports
- Analyze productivity
- Review critical employees

2. System Administrator

The System Administrator manages system operations and employee records.

Responsibilities:

- Maintain database
- Manage employee records
- Configure system settings
- Ensure system security

3. Employee

Employees are monitored based on attendance and performance metrics.

Responsibilities:

- Complete assigned tasks
- Maintain attendance
- Improve productivity

Main Use Cases

1. Monitor Employee Performance

The system tracks employee productivity and wellness continuously.

2. Generate Risk Alerts

The system generates notifications for critical employees.

3. Analyze Productivity

Machine Learning algorithms analyze employee efficiency and work performance.

4. View Employee Reports

HR managers can view detailed analytical reports.

5. Track Attendance

Attendance monitoring helps identify absenteeism patterns.

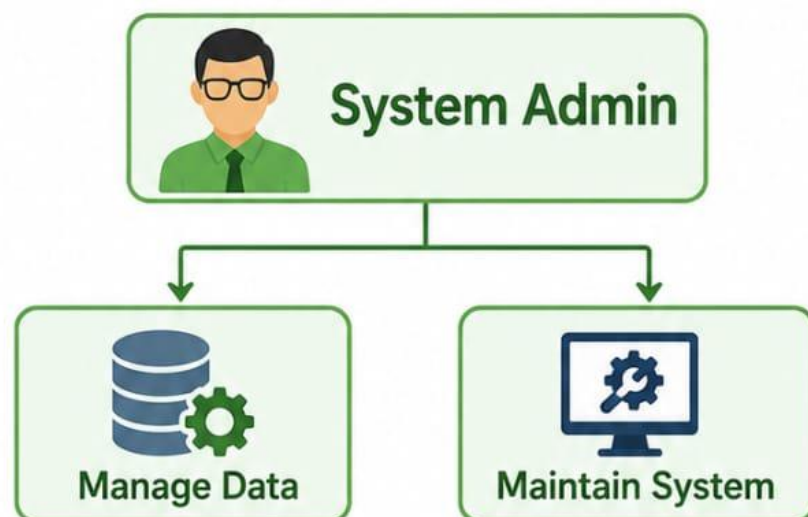
6. Reward Healthy Employees

High-performing employees receive rewards and incentives.

7. Classify Employee Risk Levels

Employees are classified into Healthy, At Risk, and Critical groups.

Use Case Diagram:



CHAPTER 7 – SYSTEM IMPLEMENTATION

7.1 List of Modules

1. Dashboard Module
2. Employee Monitoring Module
3. Risk Analysis Module
4. Alert Management Module
5. Bonus and Increment Module
6. Report Generation Module

7.2 Module Description

Dashboard Module:

Displays employee statistics including healthy employees, at-risk employees, critical employees, and bonuses awarded.

Employee Monitoring Module:

Tracks employee activities including attendance, task completion, overtime, and complaints.

Risk Analysis Module:

Calculates burnout scores based on predefined weights.

Alert Management Module:

Generates alerts for employees entering critical risk zones.

Bonus and Increment Module:

Provides incentives to healthy employees.

Report Generation Module:

Generates reports for HR management.

7.3 Algorithms

Burnout Score Algorithm

The burnout score is calculated using weighted parameters:

- Late submissions
- Absenteeism
- Average working hours
- Overtime
- Complaints
- Task completion rate

The system generates a percentage score between 0 and 100.

Classification Logic

- 0 – 39 : Healthy
- 40 – 69 : At Risk
- 70 – 100 : Critical

Formula Representation

Burnout Score = Weighted Sum of Employee Metrics

CHAPTER 8 – TESTING

8.1 Data Preparation

Employee data is collected and formatted into structured datasets.

- Data Fields
- Employee Name
- Department
- Attendance
- Task Completion Rate
- Complaints
- Overtime
- Salary

8.2 Model Evaluation

The model performance is evaluated using classification metrics.

8.3 Accuracy

Accuracy measures the percentage of correct predictions.

8.4 Precision

Precision measures how many predicted burnout cases are actually correct.

8.5 Recall

Recall measures how effectively the model identifies actual burnout cases.

8.6 F1 Score

F1 Score combines precision and recall.

8.7 Cross Validation

Cross-validation improves model reliability and prevents overfitting.

8.8 Edge Case Testing

The system is tested for:

- Missing values
- Extreme overtime
- High absenteeism
- Incorrect inputs

CHAPTER 9 – CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

- The Employee Burnout Early Detection System successfully automates employee wellness monitoring using Machine Learning concepts. The system helps organizations identify employees who are under stress or performing poorly.
- The project reduces manual HR workload, improves transparency, and supports data-driven decision-making. The dashboard interface provides clear visual insights into employee productivity and wellness.
- Overall, the project demonstrates how Machine Learning can improve workforce optimization and organizational efficiency.

9.2 Future Enhancements

Future improvements include:

1. Integration with cloud databases.
2. Real-time AI analytics.
3. Employee counseling recommendations.
4. Mobile application support.
5. Facial emotion detection.
6. Voice stress analysis.
7. Integration with biometric systems.

8. Predictive resignation analysis.

9. Advanced Deep Learning models.

10. Multi-company support.

APPENDIX

Source Code

Short Version:

<script>

```
const employees = [  
  
  {  
    id:1,  
    name:'Sarah Chen',  
    dept:'Engineering',  
    lateSubmissions:2,  
    absences:1,  
    avgHours:8,  
    overtime:2,  
    complaints:0,  
    taskRate:95,  
    salary:85000  
  },  
  
  {  
    id:2,  
    name:'Marcus Johnson',  
    dept:'Engineering',  
    lateSubmissions:8,  
    absences:6,  
    avgHours:11,  
    overtime:15,  
    complaints:2,  
    taskRate:72,  
    salary:78000  
  },  
  
  {  
    id:3,  
    name:'Emily Watson',  
    dept:'Marketing',  
    lateSubmissions:1,  
    absences:2,  
    avgHours:8,
```

**overtime:3,
complaints:0,
taskRate:92,
salary:72000
},**

**{
id:4,
name:'David Kim',
dept:'Sales',
lateSubmissions:12,
absences:9,
avgHours:12,
overtime:20,
complaints:4,
taskRate:58,
salary:75000
},**

**{
id:5,
name:'Lisa Park',
dept:'HR',
lateSubmissions:3,
absences:2,
avgHours:9,
overtime:5,
complaints:0,
taskRate:88,
salary:68000
},**

**{
id:6,
name:'James Wilson',
dept:'Engineering',
lateSubmissions:5,
absences:4,
avgHours:10,
overtime:8,
complaints:1,
taskRate:78,**

```

    salary:81000
  }

  ];

  function calculateBurnout(emp){

    let score = 0;

    score += emp.lateSubmissions * 3;
    score += emp.absences * 4;
    score += emp.overtime * 2;
    score += emp.complaints * 5;
    score += (100 - emp.taskRate);

    if(emp.avgHours > 8){
      score += (emp.avgHours - 8) * 4;
    }

    if(score > 100){
      score = 100;
    }

    return score;

  }

  function getStatus(score){

    if(score >= 70){
      return "Critical";
    }

    else if(score >= 40){
      return "At Risk";
    }

    else{
      return "Healthy";
    }

  }

```

```

function getBonus(score){

    if(score <= 30){
        return "15% Bonus";
    }

    else if(score <= 50){
        return "5% Bonus";
    }

    else{
        return "No Bonus";
    }

}

employees.forEach(emp=>{

    let score = calculateBurnout(emp);

    let status = getStatus(score);

    let bonus = getBonus(score);

    console.log("=====");

    console.log("Employee Name : " + emp.name);

    console.log("Department   : " + emp.dept);

    console.log("Burnout Score : " + score);

    console.log("Status       : " + status);

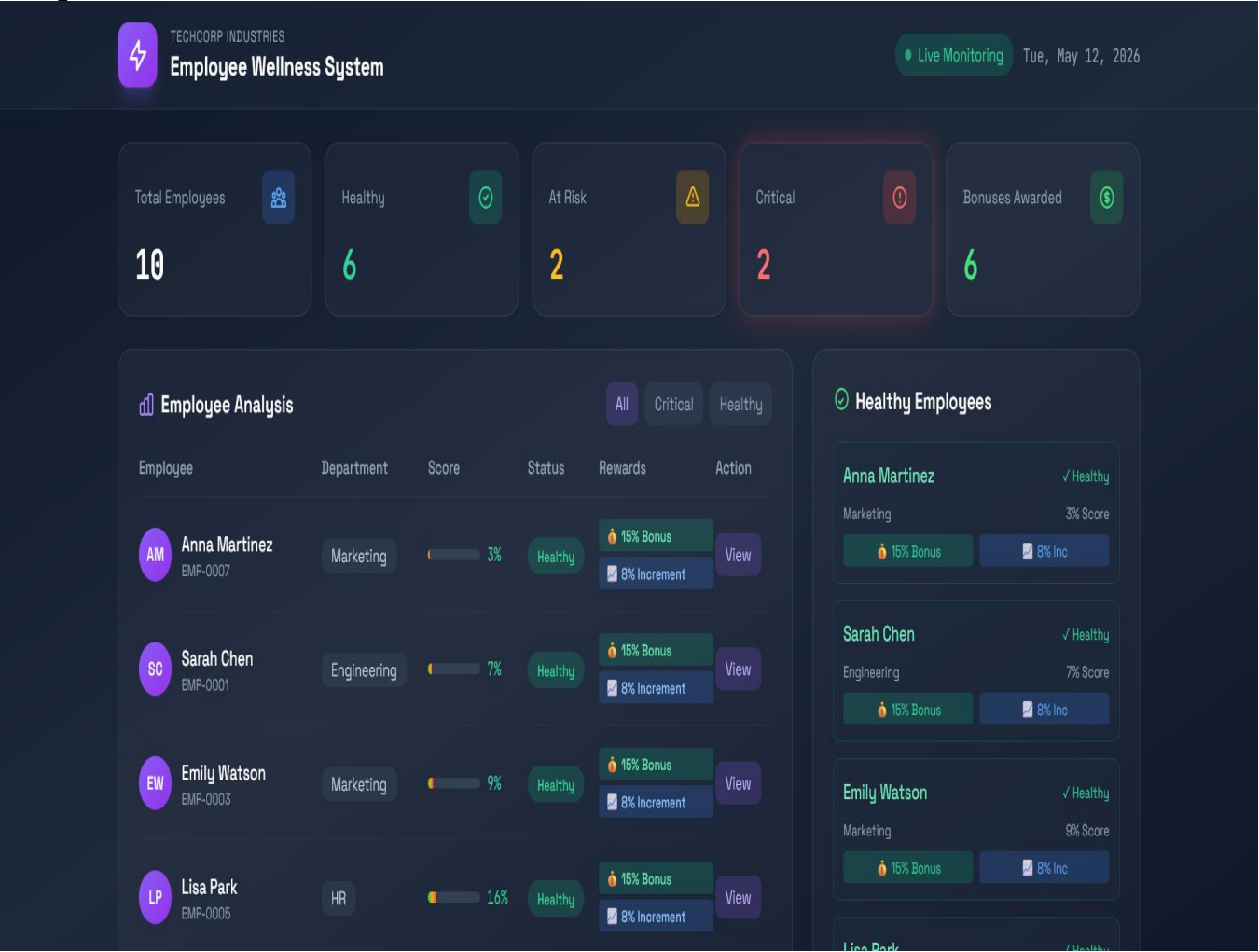
    console.log("Bonus        : " + bonus);

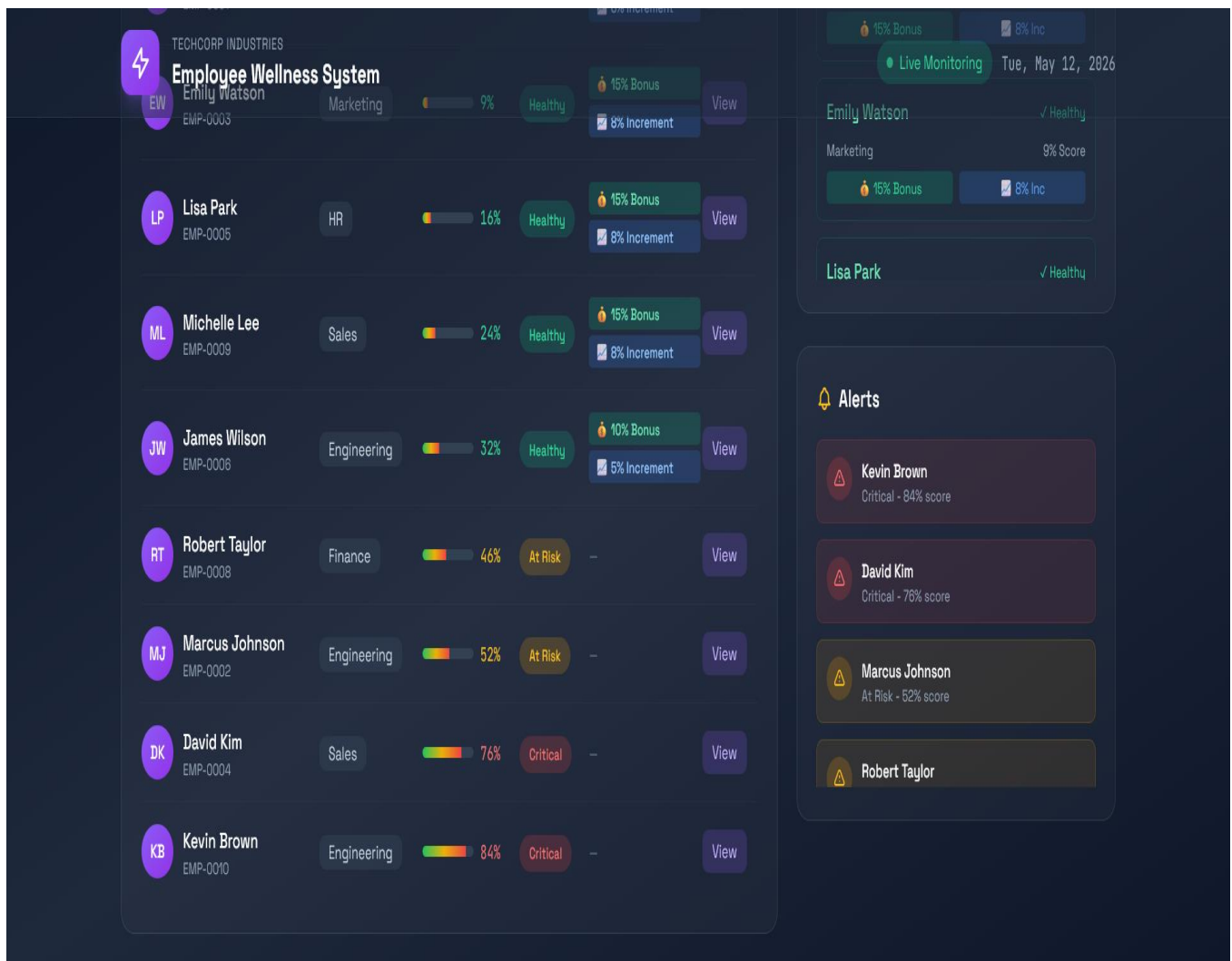
    });

</script>

```


Output:





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